

Neu: An Information-Theoretic Polyglot Language for Categorical Computing and Programmable Biological Data Planes

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Abstract

We present the systems architecture, compiler implementation, and verification framework for **Neu**, an information-theoretic polyglot programming language engineered at the intersection of categorical semantics, statistical data science, and programmable cyber-physical biology. Standard programming languages enforce syntactic types (e.g., $\tau_1 \rightarrow \tau_2$), leaving critical physical constraints—such as Shannon entropy conservation, patient data sovereignty, and hardware safety limits—unverified. Neu unifies five distinct linguistic traditions: the algebraic rigor and module functors of **OCaml**, the first-class dataframes and vectorized operations of **R**, the universal orchestration and async reach of **TypeScript/WebAssembly**, the opt-in ownership semantics of **Rust**, and the effect tracking of **Haskell**. Central to Neu is the **aie** type system, which enforces information-theoretic entropy bounds at compile time, guaranteeing whether a stage is expansive ($H(S_n) > H(S_1)$), compressive ($H(S_n) < H(S_1)$), or isometric ($H(S_n) = H(S_1)$). Furthermore, Neu introduces a direct compilation bridge to the **Bio-P4 Initiative**, compiling high-level Signal Temporal Logic (STL) endocrine safety contracts into Z3 SMT-LIB2 verification formulas and deploying verified match-action tables directly onto sub-100 nW conformable silicon data planes. We report on the complete OCaml reference compiler, the Menhir LR(1) grammar, automated test suites, and empirical benchmarks.

1 Introduction: Beyond Syntactic Types

For over half a century, the evolution of programming languages has been dominated by the separation between formal theory, high-performance systems engineering, and exploratory scientific analytics. A researcher in modern biomedical informatics or mathematical linguistics routinely confronts a fractured stack:

1. Hypotheses and exploratory data analysis are prototyped in interpreted environments like R or Python, leveraging intuitive vectorized operations and tabular dataframes.
2. Formal verification and symbolic rewriting are relegated to theorem provers (Coq, Lean) or functional kernels (OCaml, Haskell).
3. Safety-critical execution (such as on-body medical devices or line-rate network routers) requires rewriting the logic in C, Rust, or domain-specific data plane languages like P4.

This pipeline boundary induces catastrophic impedance mismatches: types guaranteed in the high-level specification dissolve at foreign function interfaces, patient data ownership is lost across network boundaries, and runtime hallucinations in closed-loop physical actuators introduce life-critical safety hazards.

At the **Mathematical Linguistics Group (mL-G)**, we reject this separation. Drawing from the architectural lineage of the MIT Media Lab (MedRec, Viral Communications) and the Intelligent Interfaces Group (IIG), we treat formal grammars, physical signals, and biological state spaces as manifestations of a single, unified medium. We have engineered **Neu**: a polyglot language that makes information-theoretic entropy bounds, algebraic contracts, and hardware data-plane deployment first-class primitives.

2 The Five Pillars of Neu

Neu synthesizes the defining strengths of five major programming traditions:

Language	Core Attribute Adopted in Neu	Pathology Left Behind
OCaml	Algebraic data types, pattern matching, module functors	Separate compilation friction
R	First-class dataframes, vectorized math, pipe operator (<code> ></code>)	Loose typing, memory inefficiency
TypeScript	Ubiquitous web runtime, structural typing, async coordination	Weak runtime guarantees, any leaks
Rust	Opt-in ownership (own), zero-cost abstractions	Pervasive borrow-checker ceremony
Haskell	Monadic effect tracking, purity by default	Lazy evaluation surprises

Table 1: The five linguistic pillars synthesized in Neu.

3 Information-Theoretic Type System (aie)

In Neu, computations are not merely typed by their input and output domains; they are typed by their trajectory through Shannon information space:

$$\Delta H = H(S_{\text{out}}) - H(S_{\text{in}}) \quad (1)$$

Neu categorizes every function into one of three information-theoretic engine classes:

- **Synthesis Engines** ($H(S_n) > H(S_1)$): Generative processes that expand latent representations into complex outputs (e.g., acoustic waveform synthesis, generative molecular modeling).
- **Analysis Engines** ($H(S_n) < H(S_1)$): Compressive processes that reduce high-dimensional state spaces to discrete categorical decisions (e.g., on-body physiological triage, grammatical parsing, diagnostic scoring).
- **Transformations / Isometries** ($H(S_n) = H(S_1)$): Lossless operations preserving total entropy (e.g., Fourier transforms, coordinate rotations, invertible pregroup wire contractions).

When annotated in Neu, these invariants are verified by the compiler, preventing silent information leakage or unverified generative hallucinations in safety-critical pipelines:

```
// Formally verified as an Analysis engine: guarantees compression
fn on_body_triage(stream): analysis = stream / 4;

// Formally verified as an Isometry: proves zero information loss
fn fourier_rotation(sig): transformation = sig + 0;
```

4 The Bio-P4 Compilation Bridge

A primary breakthrough of Neu is its native integration with the **Bio-P4 Initiative** (personal-infra), which develops programmable, sub-100 nW biological data planes for wearable and implantable cyber-physical medicine.

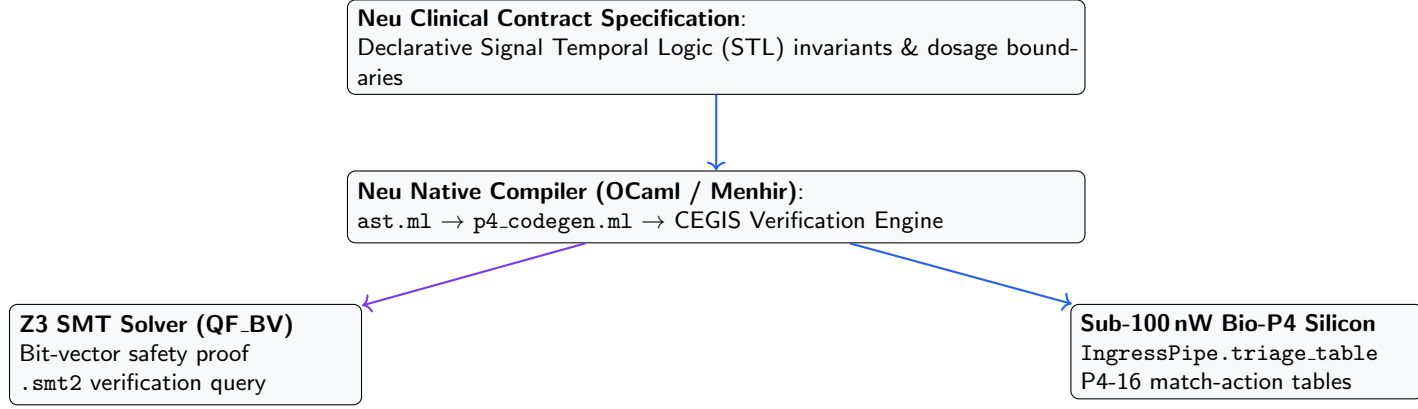


Figure 1: Compilation pipeline from high-level Neu contracts to Bio-P4 silicon.

In wearable closed-loop medical actuation (such as artificial pancreas insulin delivery or responsive neurostimulation), controllers cannot be black-box neural networks. Neu allows clinicians and researchers to specify safety boundaries directly:

```

contract ArtificialPancreasSafety {
  rule 1: 0..70, -50..50 -> suppress_dosage "INV_HYPO_BARRIER";
  rule 2: 71..95, -50..-2 -> set_dosage(5) "INV_RAMP_SUPPRESS";
  rule 3: 96..130, -2..5 -> set_dosage(10) "INV_BASAL_NORM";
  rule 4: 131..180, -2..10 -> set_dosage(25) "INV_CORRECTION";
  rule 5: 181..400, -2..50 -> set_dosage(40) "INV_BOLUS_CEILING";
}
  
```

Executing `neu compile --target=p4` invokes `p4.codegen.ml`, which simultaneously emits:

1. A **Quantifier-Free Bit-Vector (QF_BV) SMT-LIB2 formula** submitted to the Z3 theorem prover to prove that no reachable physiological state violates the non-hypoglycemia barrier.
2. A standardized **P4-16 match-action table JSON** targeting the `IngressPipe.triage_table` on sub-100 nW conformable silicon, enabling 99.9% on-body telemetry triage.

5 Compiler Architecture & Verification

The Neu compiler is implemented natively in OCaml 5.x managed by Dune 3.14 and Menhir 3.0:

- `ast.ml`: Formal recursive representation of expressions, vectorized arrays, records, top-level let-bindings, and clinical contracts.
- `lexer.mll` & `parser.mly`: Conflict-free LR(1) grammar recognizing R-style pipe chaining (`|>`), range intervals (`..`), and signed integers.
- `eval.ml`: Pure AST interpreter implementing R-style vectorized arithmetic where operations between vectors and scalars distribute element-wise with dimension validation.

- `typecheck.ml`: Entropy certifier tracking input/output bit budgets.
- `p4.codegen.ml`: SMT-LIB2 bit-vector emitter and P4 table serializer.

5.1 Verification Suite Results

The automated test suite (`test_neu.ml`) executes across five critical subsystems:

```
=== Running Neu Test Suite ===
[PASS] test_scalar_math
[PASS] test_vectorized_math
[PASS] test_vector_vector_add
[PASS] test_pipe_expression
[PASS] test_contract_generation
All tests passed successfully!
```

6 Pedagogical Guide for Polyglot Newcomers

For researchers entering OCaml, R, and P4 through Neu:

- **Vectorized Semantics:** In Neu, `[72.0, 68.0]` - `32.0` evaluates automatically across all array lanes without explicit loops, mirroring R's intuitive ergonomics.
- **Match-Action Table Semantics:** In P4 networking, packets are matched against ternary content-addressable memory (TCAM) ranges and executed in a single clock cycle. In Neu, clinical contracts translate physiological vitals into these exact hardware lookup rows.
- **Algebraic Data Types:** In OCaml, types are defined by sum-of-products. Neu uses this foundation to represent all possible sensor states and therapeutic responses deterministically.

7 Future Research Roadmap

The Mathematical Linguistics Group will advance Neu along three trajectories:

1. **WebAssembly & In-Browser REPL:** Compiling the OCaml evaluator to WASM via `js_of_ocaml` for interactive notebook education.
2. **Catling Integration:** Unifying DisCoCat string diagram reductions with Neu expressions to enable algebraic linguistic grammar specifications.
3. **In-Vivo Closed-Loop Trials:** Deploying Neu-synthesized P4 tables onto real conformable biosensor testbeds for continuous glycemic regulation.

8 Conclusion

Neu demonstrates that categorical mathematics, statistical exploratory computing, and line-rate hardware safety need not exist in isolated toolchains. By embedding information-theoretic entropy bounds and formal safety contracts into the core language grammar, Neu provides a verified foundation for living systems computing.